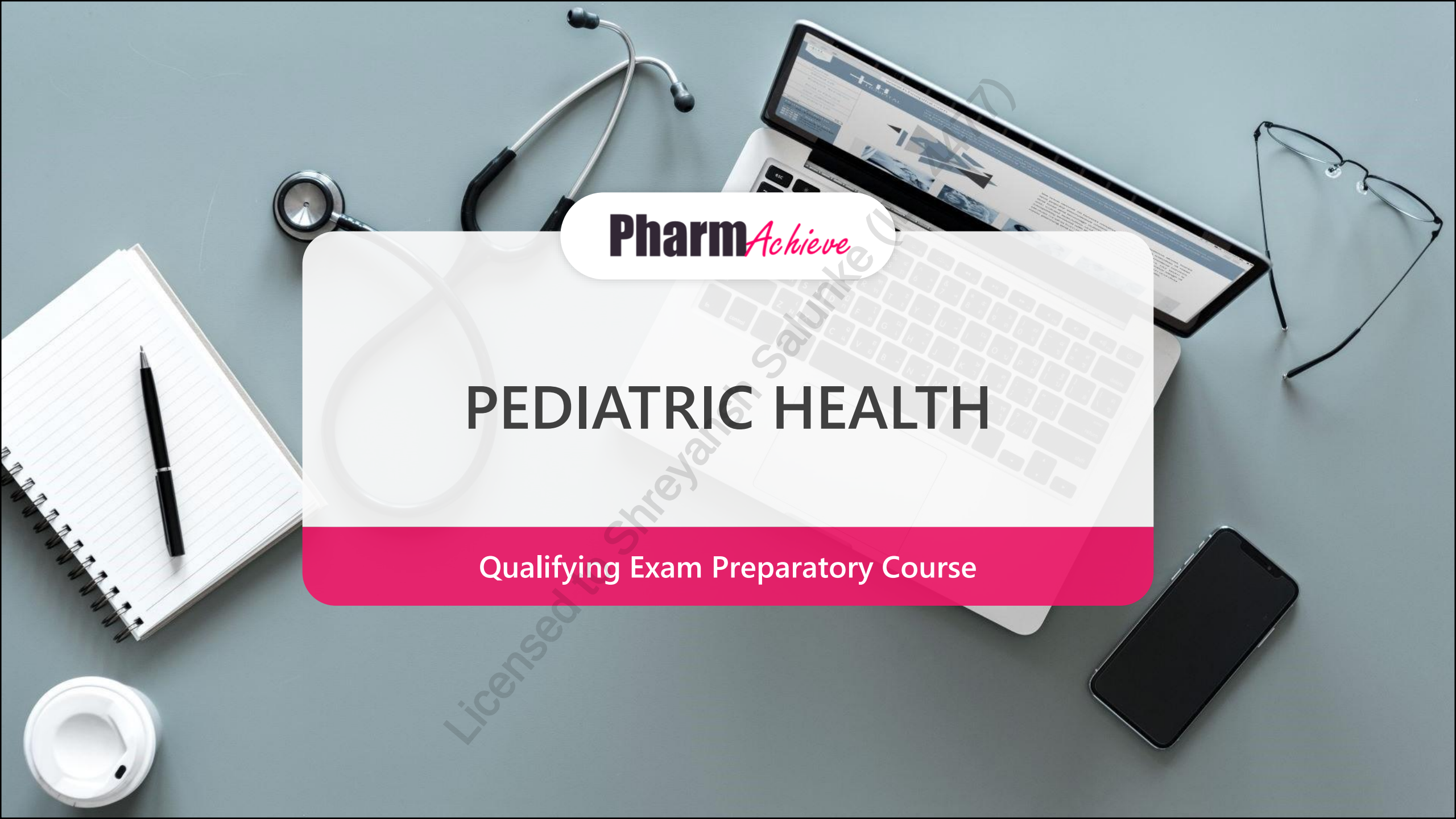


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

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PEDIATRIC HEALTH

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COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADH Anti-Diuretic Hormone

AGM Absorbent Gelling Material

BUN Blood Urea Nitrogen

Cl⁻ Chlorine

ECF Extracellular Fluid

F&DA Food and Drugs Act

GERD Gastroesophageal Reflux Disease

GIT Gastrointestinal Tract

HCO₃⁻ Bicarbonate

H₂O Water

HIV Human Immunodeficiency Virus

ICF Intracellular Fluid

IU International Units

K⁺ Potassium

KOH Potassium hydroxide

Na⁺ Sodium

NGT Nasogastric Tube

ORT Oral Rehydration Therapy

PEG Polyethylene Glycol

TBW Total Body Water

UTI Urinary Tract Infection

YR Year

OBJECTIVES

1

- Understand the following related to each condition covered:
 - Pathophysiology
 - Clinical presentation and red flags
 - Risk factors
 - Diagnosis
 - Non-pharmacological and pharmacological therapy
 - Monitoring parameters

2

Compare therapeutic options in terms of efficacy and adverse effects

INFANT NUTRITION

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INFANT NUTRITION AND SUPPLEMENTATION

The Canadian Pediatric Society and Health Canada recommend to exclusively breastfeed infants for the first 6 months, after which solid foods can be introduced

- Breastfeeding can be continued for up to 2 years with appropriate complementary feeding
- Very few contraindications to breastfeeding (consider alternatives in galactosemia, maternal infections such as HIV, chemotherapy...etc.)
- Supplementation with **Vitamin D is recommended in infants who are being breastfed**
 - Vitamin D 400 IU daily for one year
 - Vitamin D 800 IU daily in certain populations (i.e. northern communities or other risk factors)
- Supplementation **with Vitamin D may not be required in formula-fed** infants (since formula contains Vitamin D) though those with certain risk factors may need supplementation

INFANT FORMULAS

Regulated by the F&DA

Cow Milk-Based Formula | Preferred option in all infants

- Contains lactose and cow's milk-based protein
- Lactose plays a role in mineral absorption and bacteria colonization in the GIT

Lactose-Free Cow Milk-Based Formula

- Contains cow's milk-based protein. Lactose is replaced with maltodextrin and sucrose
- Considered in infants who present with gas or are fussy due to the lactose content in cow milk-based formula

Soy Protein Isolate-Based Formula

- Does not contain lactose or milk protein (contains soy isolate)
- Recommended in infants born at term who have galactosemia or congenital lactase deficiency

Hydrolyzed Protein Formula | Dependent on the degree of filtration

- Partially hydrolyzed: contain lactose and whey protein
- Extensively hydrolyzed: contains casein protein and recommended in intolerance to soy protein and intact cow milk

Amino Acid-Based Formula

- Contain free amino acids, lactose-free
- Recommended in severe milk protein hypersensitivity

INFANT COLIC

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BACKGROUND

What is colic?

- Persistent and excessive crying for no apparent reason
- Although it is self-limiting, it can negatively impact parents
- Self-resolving: symptoms spontaneously improve with time, with resolution generally by 3 to 4 months of age

How does colic affect families?

- Families with colicky infants have been shown to have more problems in family structure, functioning, stress, parental self-confidence and affective state than control families

Pathophysiology is not completely understood. Possible causes include:

- **Gastrointestinal** (e.g. lactose intolerance, GERD, intestinal gas, cow's milk protein intolerance, feeding techniques). Evidence suggesting that changes in the intestinal microbiome may play a key role, where colicky infants may have altered microbial patterns (e.g. impaired or delayed colonization with *Lactobacillus species*)
- **Psychosocial** (e.g. maternal anxiety, infant exposure to substances e.g. nicotine, smoking)

HOW TO DIFFERENTIATE BETWEEN COLIC AND NORMAL CRYING

Paroxysms	Qualitative differences	Difficulty consoling	Hypertonia
• Sudden onset of episodes • Often cluster during the evening hour • Spells of crying lack a clear trigger	• Louder, higher in pitch • More turbulent and dysphonic • Sounds as if the infant is in pain instead of crying	• Infants cannot be consoled no matter what the parents do • Crying may diminish although the infant remains fussy	• Characteristics of hypertonia may present during episodes • Flushing, circumoral pallor, tense or distended abdomen, arched back, clenching of fingers



These clinical features occur in non-colicky infants as well.
Use patient history and Wessel/Rome IV criteria to assess

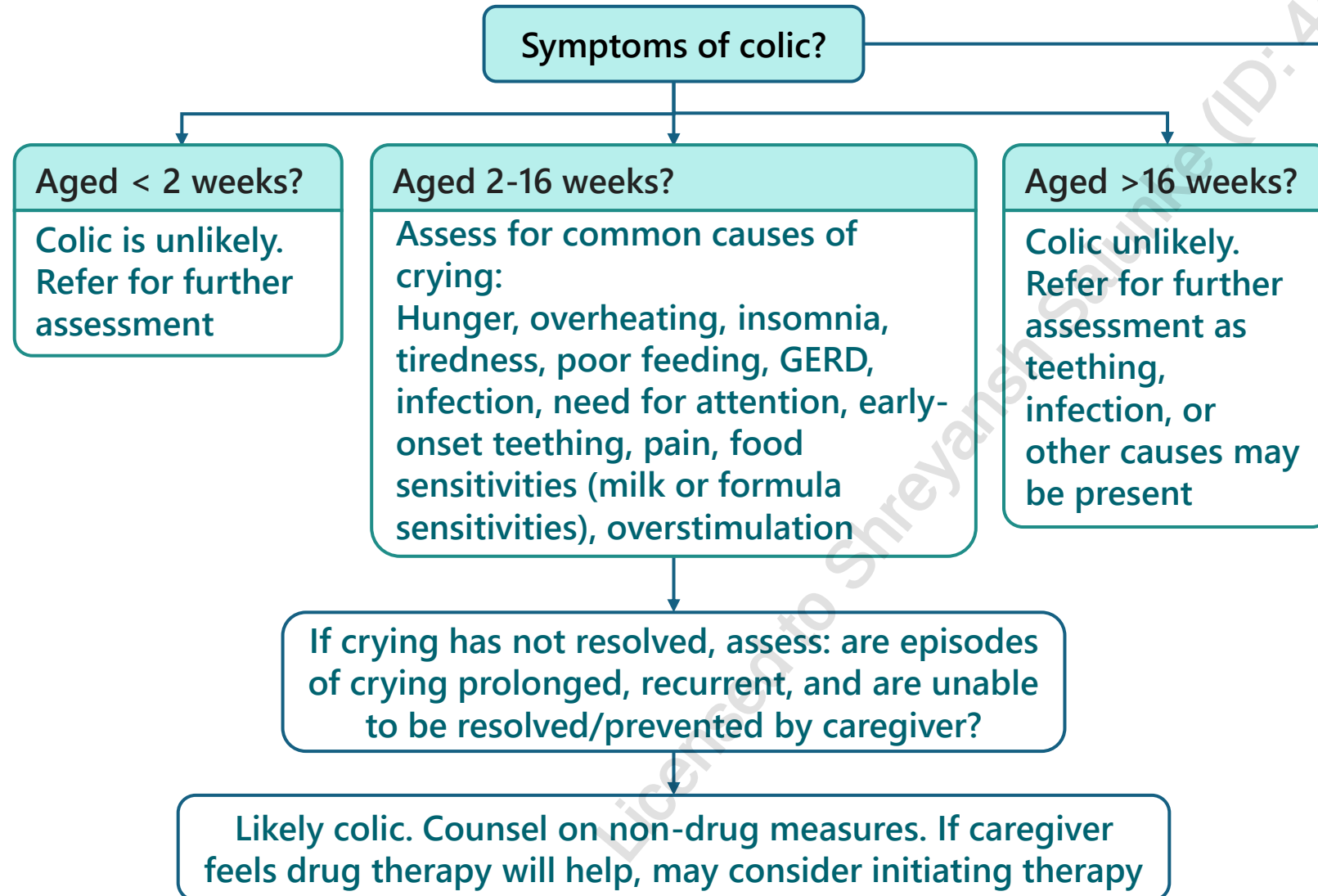
HOW IS COLIC DIAGNOSED?

FYI

Wessel's criteria "Rule of Three"	Rome IV criteria
• ≥ 3 hours per day • ≥ 3 days per week • Persists for ≥ 3 weeks • Patient healthy and well fed • To be more specific, also satisfy 3 of the following: • Paroxysmal, qualitatively different from normal crying, hypertonia, inconsolability	• Age < 5 months • Recurrent and prolonged periods of crying that cannot be prevented or resolved • No evidence of failure to thrive, fever, or illness

Note, there is no standard definition of colic!

TREATMENT ALGORITHM



Any one of the following red flags requires referral:

- Continuous crying for >3 hours
- ≥ 4 months with excessive crying or age >5 months
- High-pitched or extreme crying
- Fever $\geq 38^{\circ}\text{C}$ (medical emergency in infants < 3 months of age)
- Lethargy
- Weight loss
- Dehydration
- Failure to thrive
- Diarrhea
- Mucous or blood in stools
- Frequent vomiting or regurgitation
- Frequent abdominal distension
- Unpredictable diurnal rhythm
- Parental mood disorders e.g. anxiety
- Maternal drug ingestion (prescribed or illicit use)
- Family history of atopy or migraines
- Any physical harm to the baby like shaking

NON-PHARMACOLOGICAL MEASURES

1st Line

Address feeding problems

- Refusing to be fed
 - Reduce the amount of air swallowed during bottle feeding by positioning the infant to be more upright
 - The provider may suggest trialing a different formula for one week, then reverting to the original if no change in colic status
 - Modify the breastfeeding mother's diet to hypoallergenic foods to reduce the infant's colic (eliminate products like milk, egg, wheat)
- Crying between feedings suggests hunger
- GERD
- Teething

Colic diary

- Helps determine behaviors and patterns

Assess their environment and stimulation

- Overheated
- Insomnia
- Need for contact or attention
- Lack of stimulation or over stimulated



- Remove clothing layers
- Consider carrying, rocking, and swaddling to soothe the baby
- If overstimulation occurs, consider a dark and quiet room
- Avoid stimulation before bedtime
- Consider distractions or playing with the infant in lack of stimulation

PHARMACOLOGICAL MEASURES

Pharmacologic therapy is rarely recommended due to lack of understanding of the pathophysiology of colic

Use OTC therapy to minimize parental frustration and consider value of placebo:

- **Carbohydrates:** *Sucrose* has proposed analgesic effects, where limited data shows a slight decrease in symptoms. However, its short duration of effect (30 minutes) limits use
 - Less evidence than probiotic agents
- **Carminatives:** *Simethicone* has anti-flatulent effects
 - It has been shown to have a placebo effect, but is still used due to its safe side effect profile. It is the only product that has been licensed by Health Canada for infant colic
- **Probiotic agents:** *Lactobacillus reuteri* DSM 17938 strain has demonstrated meaningful and significant reductions in crying times
 - Colicky infants have lower counts of intestinal *lactobacilli*

MONITORING PARAMETERS

Monitoring Parameter	Outcome
Excessive crying frequency or duration decreased	After 3 days or at next visit
Minimized post-feed fussiness or discomfort	
Improvement in sleep	
Decreased stress or anxiety in caregiver	

TEETHING

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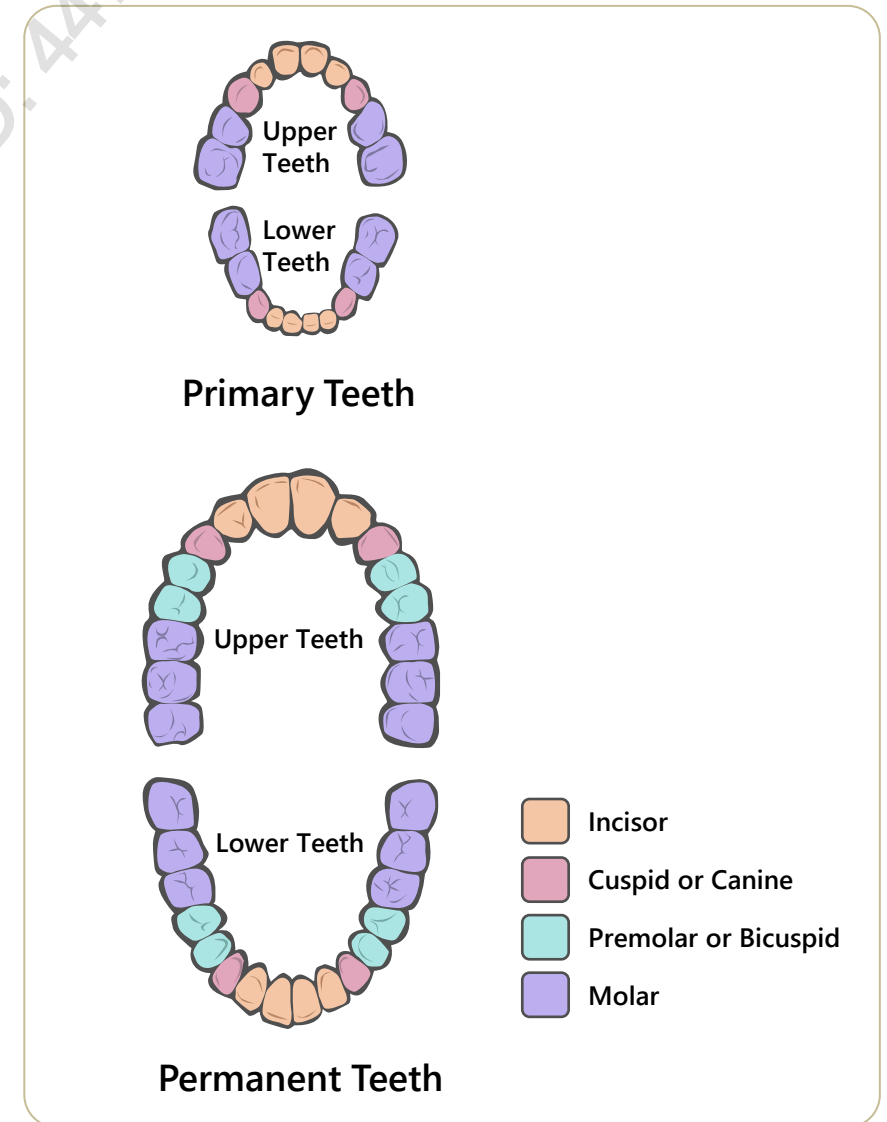
BACKGROUND AND PATHOPHYSIOLOGY

Teething: When teeth emerge from the gums

- Begins around 6-7 months of age
1. **Primary Teeth** (*by 2-3 years of age*)
 - Eruption is associated with **symptoms** in many infants
 2. **Permanent Teeth** (*at 6 or 7 years of age*)
 - Eruption is associated with **less pronounced symptoms**

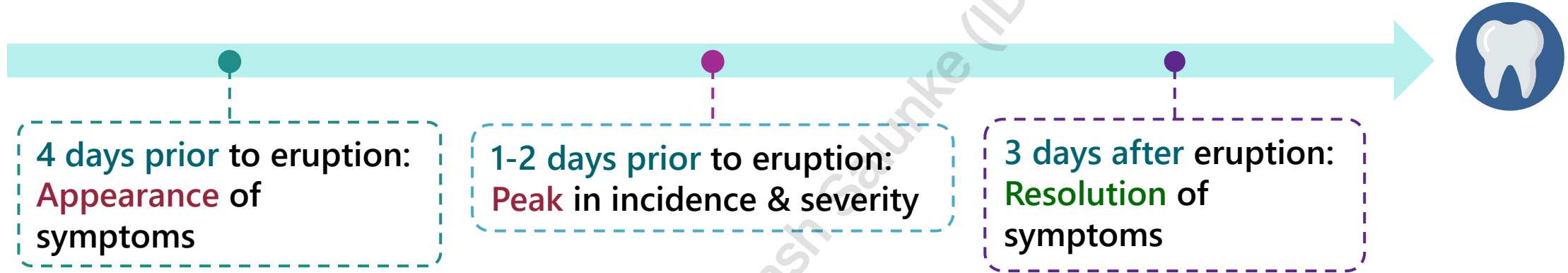
Delayed Tooth Eruption

- Delays up to **6 months and 12 months** for primary and permanent teeth, respectively **is normal**
- Several causes for delayed tooth eruptions:
 - **Systemic causes** (e.g. Down syndrome)
 - **Local causes** (e.g. cysts)
 - **Iatrogenic causes** (e.g. chemotherapy)



CLINICAL PRESENTATION

Timeline of Primary Teeth Eruption (Symptom Severity)



- Gum overlying the tooth will show signs of:
 - **Inflammation, redness, irritation, swelling, tenderness, and pain**
- Local inflammation can manifest as:
 - **Agitation, restlessness, crying, loss of appetite, increased thirst, and mild fever**
- Child will have a greater tendency to rub the gums and bite on fingers, lips and toys
 - This induces **salivation and drooling (causing facial irritation and skin rash)**

RED FLAGS AND WHERE TO REFER



Possible systemic illness	e.g. fever, diarrhea, vomiting...etc.	→	Medical referral
Pericoronitis	severe inflammation and swelling	→	Dental referral
Pericoronal abscess	pericoronitis with pus	→	Urgent dental or urgent medical referral
Delayed tooth eruption	more so than normal	→	Dental referral
Eruption cysts	with no spontaneous drainage	→	Dental referral

NON-PHARMACOLOGICAL MEASURES

Teether

- Usually a **hard, smooth and clean object** given to a child to bite and chew on to minimize discomfort (e.g. firm rubber teether) - make sure not too hard, can hurt gums
- **Cool in refrigerator** before use and ensure no small parts—make sure not frozen
- **Avoid long-term contact** with very cold items (concerns of **hypothermia**)
- Supervise to avoid choking

Rub the gums

- Using a **clean finger or cold spoon**, caregivers can gently rub the gums to minimize discomfort

Wipe away drool

- To minimize facial irritation, ensure that the **face is kept clean and dry**



- Teething biscuits are **NOT** recommended due to high sugar content
- Amber teething necklaces are **NOT** recommended due to risk of strangulation and lack of clinical benefit

PHARMACOLOGICAL THERAPY

Oral Analgesics

Acetaminophen

- Children: 10 to 15 mg/kg/dose every 4 to 6 hours by mouth as needed
- Maximum daily dose: 75 mg/kg

Ibuprofen

- Children <6 months: 5 mg/kg/dose by mouth every 8 hours
- Children ≥6 months: 5 to 10 mg/kg/dose every 6–8 hours by mouth as needed
- Maximum ibuprofen daily dose: 40 mg/kg



Topical anesthetics (e.g. benzocaine) are NOT recommended to be used for teething in children <2 years old due to the risk of methemoglobinemia if consumed

DIAPER DERMATITIS

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BACKGROUND AND PATHOPHYSIOLOGY

Also known as diaper rash

Clinical presentation

Gradual onset of symptoms present only in the diaper area, can appear moist with shiny patches on convex areas (i.e. buttocks), patches of erythema and scaling may be present

Main types

- Irritant diaper dermatitis
- Candida diaper dermatitis
- Allergic diaper dermatitis

Common causes: diaper via increased friction, excessive moisture and chemical exposure

Increased skin pH: alkaline urine and feces increases skin pH, making it more susceptible to organism colonization (e.g. *Staphylococcus aureus*, *Candida albicans*)

Skin irritants: activated fecal enzymes (e.g. protease, lipase) directly damage skin

Diet: changes in fecal bacteria

Yeast: candida infections (prevalent in moist environments and intestinal flora)



All of these contribute to a disruption in function of the skin barrier

CLINICAL PRESENTATION

Irritant Diaper Dermatitis

- **Most common type** of diaper dermatitis that presents as a **chafing rash** where the **skin folds are unaffected**
- Can appear deep-red or purple on darker skin
- Severity can increase or decrease with time and can spontaneously self-resolve

Candida Diaper dermatitis

- Second most common type of diaper dermatitis
- *Patients who have untreated diaper dermatitis for > 3 days are at a higher risk of secondary infection with Candida albicans infection*
- Infection severity correlates with severity of existing dermatitis
- Described as a **"beefy red rash,"** usually **involving the skin folds**
 - Presents as diffuse erythematous patches that will progress to bright red plaques, papules and pustules that will appear to merge together with scaling on the peripheral border
- Presence of **oral thrush** may be seen

RED FLAGS



- Moderate-severe symptoms or complicated diaper dermatitis (oozing, blisters, systemic symptoms, worsening symptoms)
- No improvement after a week of treatment
- Does not resolve after 2 weeks of treatment
- Symptoms spreading past diaper area (regardless of treatment)
- Recurrent dermatitis (especially when there is no rash-free period)
- Signs of neglect or abuse
- Presence of secondary infection or UTI
- Immunodeficiency
- Significant behavioural changes (lethargy, incessant crying)

DIAGNOSIS

Physical examination and patient history

- Clinical presentation
- Rule out differential diagnosis
- Patient history (systemic symptoms, concurrent symptoms, information about diapering, history of infection or inflammatory skin disease, risk factors...etc.)

Laboratory

- Generally, not required in irritant dermatitis, however might be useful in resistant or atypical cases
- Consider KOH and fungal culture in resistant candida diaper dermatitis (to confirm *Candida* superinfection)

PREVENTION – ABCDS

AIR AND ABSORPTIVE MEASURES

A

AIR

- Always let the area air dry
- Do not dry the area using a hair dryer or an infrared lamp
- Use breathable materials

Absorptive

- Zinc oxide is effective at absorbing moisture at higher concentrations (up to 40%)
- Avoid cornstarch as it can promote fungal growth

BARRIER MEASURES

B

Zinc oxide

- Acts as a barrier, antiseptic, astringent and absorptive (at higher concentrations)
- **Prevention** use in concentrations of **10-20%**, **treatment** use in concentrations of **20-40%**

Petrolatum

- Oil based and very effective at trapping moisture
- Can be irritating to inflamed skin and there is risk of maceration if applied to overhydrated skin

Barriers are a physical shield, remain intact, and maintain the skin's acidic pH, and should be applied with each diaper change

PREVENTION – ABCDS

C

CLEANSING MEASURES

- Rinse the area with warm water and gently cleanse using fragrance-free, preservative-free, alcohol-free baby wipe or a soft cloth
- For fecal removal, mild soaps can be used and for urine removal, water is sufficient
- Keep the area clean and dry with frequent diaper changes if needed
- Diaper wipes can be used on intact skin as long as they don't contain irritating **chemicals like alcohol, fragrances, lanolin, methylisothiazolinone, soaps**

D

DIAPERS MEASURES

Disposable Diapers

- Contains an absorbent polymer core with a breathable outer cover, known as AGM diapers
- **AGM diapers are able to absorb moisture and remove it away from the skin in a vertical manner; reducing mixing of urine and feces as well as maceration**
- Do not double-diaper as it is not proven to be effective
- Do not clean skin using unsoiled portion of diaper

Cloth Diapers

- Not as effective at reducing moisture compared to AGM diapers
- Necessary to wash them properly
 - ✓ Wash with mild detergent
 - ✓ Add a cup of vinegar in the final rinse to reduce the pH
 - ✓ Preferable to have them professionally washed by a diaper service as they can sterilize them

NON-PHARMACOLOGICAL

Compress

- Indicated for dermatitis that is oozing and/or crusting along with inflammation
- Compresses dry and cool the skin, reduce inflammation, cleanse the skin, drain infected areas
- How to properly apply compresses:
 - Soak a clean piece of cloth or non-irritating gauze with solution (tap water, normal saline, or astringent)
 - Wring it so that it is no longer dripping and apply to the skin
 - Remove the cloth, re-soak and apply again every few minutes for up to 30 minutes x 4-6 times a day
 - Do not use wet soaks as it will macerate and harm the skin

PHARMACOLOGICAL

Zinc oxide should be increased to 20-40%

Topical Antifungals (in suspected Candidal infection or perineal irritation persisting for 3 or more days)

- First line options: ketoconazole 2%, miconazole 2%, clotrimazole 1% (apply BID to TID for a minimum of 1-2 weeks; improvement can be seen in 1 week)
- Nystatin is slightly less efficacious (apply BID to TID for at least 1 – 2 weeks (usually takes 2 weeks to see improvement)
- Ciclopirox is superior to nystatin but inferior to imidazole (apply BID for a minimum of 4 weeks). Safety not established in those <10 years

Topical Corticosteroids

- Hydrocortisone 0.5-1% is recommended for severe inflammation
 - Apply up to BID PRN for up to 2 weeks
 - Do not use high-potency corticosteroids due to risk of drug-induced skin atrophy

Do not mix equal parts of an antifungal + corticosteroid as the resultant concentration will be halved



MONITORING PARAMETERS

Monitoring Parameter	Outcome
Symptom improvement (80% reduction in inflammation)	Improve in 1 week and resolve in 2 weeks No new blisters after initiating treatment No more surface area involvement after initiating treatment
Behavioral changes return to baseline	In 2-3 weeks
Medication safety	Most topical therapies are well-tolerated, monitor for any worsening dryness or irritation

DEHYDRATION IN CHILDREN

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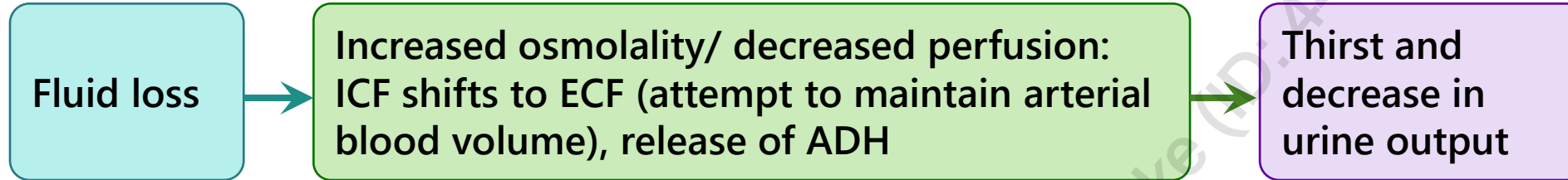
INTRODUCTION

Dehydration occurs when water loss is greater than water intake

Hypovolemia is characterized by volume depletion secondary to salt and/or water loss

Dehydration, sodium loss, and hypovolemia may be related to, or independent of, each other

PATHOPHYSIOLOGY



CAUSES OF FLUID LOSS

GIT causes

- Gastroenteritis, diarrhea, vomiting

Dermatological causes

- Fever, burns, excessive sweating

Gastro-urinary causes

- Diabetes insipidus
- Diabetes mellitus
- Diabetic ketoacidosis
- Diuretic use

Inadequate intake of fluids or malnutrition

RISK FACTORS FOR FLUID LOSS

High surface area to volume ratio

- Children 60-80% TBW whereas adolescents/adults 55-60% TBW

Increased metabolic rate

Difficulty or inability to communicate

Higher risk of viral gastroenteritis

- Rotavirus
- Norovirus
- Cholera

Age! Children are at risk (the lower the age, the greater the risk)

SODIUM IMBALANCE

	Hyponatremia	Isonatremia	Hypernatremia
Definition	Loss of Na^+ > Loss of H_2O = loss of Na^+	Loss of Na^+ = loss of H_2O	Loss of Na^+ < Loss of H_2O = excess Na^+
Serum Na^+	< 130 mmol/L	130-150 mmol/L	>150 mmol/L
Symptoms	Nausea, vomiting, seizures, neurologic damage, coma, death	Rapid pulse, increase respiration, decreased bp, shock, lethargy	Excessive thirst, lethargy, irritable, tremors, seizures, coma
Comments	• Least common type of dehydration (5%) • Causes: severe diarrhea and vomiting, rehydration with plain water, burns, diuretics	• Most common type of dehydration (80%) • Causes: diarrhea, vomiting, malnutrition, gastroenteritis	• Second most common type of dehydration (15%) • Causes: diabetes insipidus or high electrolyte rehydration

CLINICAL PRESENTATION

Dehydration	Mild	Moderate	Severe
Volume Loss*	3-5%	6-9%	≥10%
Clinical signs and symptoms	Little or no signs and symptoms: • Slightly thirsty • Slight decrease in urine output • Capillary refill <2 s (normal)	• Thirsty • Decrease in urine output • Decreased tear production • Dry mucous membranes • Slight increase in pulse • Normal to lowered BP (by 10 mmHg) • Irritable behaviour • Sunken eyes • Normal to sunken fontanelle • Reduced skin turgor	• Very thirsty • Decrease or no urine output • Absence of tears • Dry mucous membranes • Rapid pulse • Hypotension • Lethargic, coma • Distinctly sunken eyes • Distinctly sunken fontanelle • Reduced skin turgor and tenting

*Children 1 year or older are at the lower end of the range, and infants under 1 year are at the upper end of the range

DIAGNOSIS

Medical history

Rule out underlying causes

Physical examination

Fluid deficit: change in urine output, total volume deficient (L) = change in weight (kg) where baseline weight often unavailable
Blood pressure, respiratory rate, pulse, temperature, mental status
Skin turgor, increased thirst, increased lethargy or irritability, decreased tear production

Lab testing

Reserved for moderate-severe cases requiring IV fluids
Test: serum electrolytes, blood glucose, BUN, creatinine, osmolarity, sodium

ORAL REHYDRATION THERAPY (ORT) AND ONDANSETRON

- First-line therapy in mild-moderate dehydration
 - Equally effective as IV fluids
 - Safer and less expensive/healthcare intensive
- Give small amounts frequently
- ORT may be administered through NGT
- PO/IV ondansetron can be used to decrease vomiting and increase tolerance to ORT
 - Reduces the need for IV fluids and hospital admission
 - ORT should begin 15-30 minutes post-antiemetic administration

Rate:

- Mild-moderate dehydration: 50 mL/kg over 4 hours
- ORT is not first-line therapy for severe dehydration, however:
 - If using in severe dehydration: 100 mL/kg over 4 hours
- After first 4 hours, replace remainder of deficit over next 6-8 hours and continue maintenance hydration

ORAL REHYDRATION THERAPY SOLUTIONS

Dosing is FYI

	DEXTROSE (g/L)	Na ⁺ (mmol/L)	K ⁺ (mmol/L)	Cl ⁻ (mmol/L)
WHO revised formula	13.5	75	20	65
Gastrolyte®	17.8	60	20	60
Pedialyte®	25	45	20	35
Hydralyte®	16	46	22	43

INTRAVENOUS FLUIDS

Includes normal saline (0.9% NaCl) and Ringer's lactate

- First-line therapy in patients with severe dehydration
 1. Normal saline IV boluses (20 mL/kg over 10-15 min) recommended to restore volume, blood pressure, and perfusion
 2. Then calculate any remaining deficit, plus maintenance (see Holliday-Segar method), and replace 50% over the next 8 hours
 - Replace final 50% over 16 hours

Body weight	Daily fluid requirements	Hourly rate
≤10 kg	100 mL/kg	4 mL/kg
11-20 kg	1000 mL + 50 mL for each kg over 10 kg	40 mL/h + 2 mL/h for each kg over 10 kg
>20 kg	1500 mL + 20 mL for each kg over 20 kg	60 mL/h + 1 mL/h for each kg over 20 kg

Note: children who have started IV fluid replacement should be de-escalated to oral intake when stable as it is equally effective and reduces hospital stays

OTHER FLUID AND ELECTROLYTE REPLACEMENTS

NaCl 0.45%

- NaCl 0.45% is used to (slowly) restore volume and Na^+ to normal levels in hypernatremic dehydration
- Shock is first treated with normal saline 0.9% NaCl

NaCl 3%

- NaCl 3% is used to correct severe/life threatening hyponatremia

D5W

- Isotonic carbohydrate solution used to restore hydration in hypernatremic dehydration
- Avoid in patients with renal failure, cardiac compromise, or those who are at risk for increased intracranial pressure

K^+

- Avoid in anuria; daily maximum is 4 mmol/kg

COMPLICATIONS OF REHYDRATION

- HYPOvolemia
- HYPERvolemia
- HYPERnatremia
- HYPOnatremia
- HYPOkalemia
- HYPOphosphatemia
- HYPERchloremia
- HYPOglycemia
- HYPERglycemia
- Mortality

THERAPEUTIC TIPS

- Continue monitoring the patient throughout therapy
- If the patient protractedly vomits or is unable to drink, then treatment can be administered via:
 - IV fluids or via NGT
- Introduce food when stable but slowly to avoid refeeding syndrome
- Do not rehydrate with plain water
 - May cause hyponatremia and hypoglycemia
- Avoid soda, sports drinks or juice as rehydration fluids
- Refer all moderate and severe cases of dehydration, patients <6 months old and patients with diarrhea >24 hours or vomiting >6 hours

TAKEAWAY

- Supplementation of Vitamin D 400 IU daily for a year is recommended in infants who are being breastfed
- Infant colic: persistent and excessive crying for no apparent reason; self limiting
 - Non-pharmacological measures are 1st line
 - Pharmacological measures to minimize parental frustration (sucrose, simethicone, probiotic)
- Teething: nonpharmacological and/or OTC simple analgesics (no topical anesthetics)
- Diaper dermatitis
 - Zinc oxide 20-40% for treatment
 - Topical antifungals in Candidal infection
 - Topical low potency steroids for severe inflammation
 - Order of application: antifungal, steroid, barrier (UNLESS antifungal is in a PEG base then apply the steroid first)
- Dehydration
 - Mild-moderate dehydration: ORT
 - Severe dehydration: IV fluids

CASE 1

CG is a 6-month-old female who was brought into the clinic by her mother. CG weighs 7.5 kg and has normal vital signs. CG has no known allergies and no significant medical history. CG's mother is concerned about a red rash in CG's diaper area that has been present for 4 days with no improvement. The rash appears to be inflamed with some small red bumps and satellite lesions. CG has been more fussy than usual and has been crying during diaper changes. CG's mother read online about how to treat diaper rash. CG's mother has tried increasing the frequency of diaper changes, air drying the area, applying zinc oxide 20% and hydrocortisone 1% cream daily.

1. Which of the following differentiates between irritant and candidal diaper dermatitis?
 - a) Irritant presents as erythema and can affect inguinal folds
 - b) Candida presents with beefy red plaques and involves the inguinal folds
 - c) Candida presents with satellite pustules and does not affect inguinal folds
 - d) Irritant can be associated with recent antibiotic use



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2. In which order should topical medications should be applied for diaper dermatitis?

- a) Steroid, barrier, antifungal in PEG base
- b) Steroid, antifungal in PEG base, barrier
- c) Antifungal in PEG base, steroid, barrier
- d) Antifungal, barrier, steroid



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3. Which of the following is the next best step for CG?
- a) Refer CG to doctor due to behavioral changes
 - b) Recommend ketoconazole 2% cream
 - c) Recommend a compress
 - d) Determine how the products have been applied



CASE 1

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4. What advice would you give CG's mother regarding the prevention of future episodes of diaper dermatitis?
- a) Consider using a hair dryer to speed up the process of drying
 - b) Consider using a compress on the area
 - c) Consider continuing zinc oxide 20%
 - d) Consider double diapering



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CHANGE LOG

April 2025

- Formatting and content changes throughout
- Slide 11 now changed to FYI
- Slide 12: Addition of content to red flags and symptoms for ages 2-16
- Slide 13: Addition of details for refusing to be fed
- Slide 23: Modification to content to reflect new information
- Slide 28: Addition of content to cleansing measures
- Slide 30: Updated first line therapy to reflect guidelines (UpToDate)
- Slide 36: Updated volume loss to reflect guidelines (UpToDate)
- Slide 37: Updated lab testing to reflect guidelines (UpToDate)

August 2025

- Updated legend
- Minor formatting changes made

October 2025

- No content changes made
- Slide 9: Simplified timeline of colic resolution

February 2026

- Slide 9: Simplified timeline of colic resolution, and added to psychosocial causes